IT-DBS1 Exam S2026

Before the exam

Run both:

- Simple_Goodreads_ddl.sql
- Simple_Goodreads_data.sql

in DataGrip, so that you have the database available when you show up for the exam.

These files will be available in Itslearning and WISEflow at the latest one week before the exam.

Also, it's recommended to have Astah installed, and working, when you show up for the exam.

Exam information

The exam is 20 minutes for each student, including grading and feedback.

You will draw one question with two parts:

- The first part focuses on one of your course assignments.
- The second part will require you to either write and explain a given SQL query using the Simple_Goodreads database, or create an ER diagram.
  You will get a small case at the exam for the second part.

Question 1 – SQL Queries

Tasks:

1. Present your **Assignment 1** and explain how you used joins, groupings, aggregate functions, and set operations. Explain some of the insights you got from the results of the queries.

2. Based on a small given case, draw an ER diagram for the situation described in the case.

Question 2 – Entity-Relationship Modelling

Tasks:

1. Present your **Assignment 2**, focusing on how you found the entities and relationships, and designed the diagram.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.

Question 3 – Enhanced ER Modelling

Tasks:

1. Present your **Assignment 2**, focusing mainly on how you added EER features and why they were needed.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.

Question 4 – Mapping EER Models to Relational Schemas

Tasks:

1. Present your **Assignment 3**, focusing on how you mapped your EER diagram into a relational schema, and your considerations about primary keys and foreign keys.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.

Question 5 – Global Relations diagrams

Tasks:

1. Present your **Assignment 3,** focusing mainly on your GR diagram, and how it is different from the ER diagram that was mapped.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.

Question 6 – Creating Databases with DDL

Tasks:

1. Present your **Assignment 4** and explain how you wrote your CREATE TABLE statements, what types, keys, and constraints you chose, and why.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.

Question 7 – Functional Dependencies and Keys

Tasks:

1. Present your **Assignment 5,** focusing mainly on which functional dependencies you found, and how they helped you choose your keys.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.

Question 8 – Normalization to 1NF, 2NF, and 3NF

Tasks:

1. Present your **Assignment 5,** focusing mainly on how you normalized your data, step-by-step from unnormalized to 3NF.

2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.